Jia-Fong Yeh (葉佳峯)

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PhD in Computer Science (NTU) specialized in **Safe and Embodied AI**. Proven track record in developing reliable autonomous systems with publications in top-tier global venues (AAAI, NeurIPS, ICLR). Experienced in leading cross-functional research teams and managing international collaborations (Sony, Delta Electronics) to drive industrial innovation.

Working Experiences

Applied Scientist II, Amazon RBKS AI *Mar. 2026 – current*

Toronto, Canada

- Start a new chapter in a cross-functional team, working on generative AI solutions on Ring products.

Senior Associate Researcher, Delta Electronics *Aug. 2025 – Jan. 2026*

Taipei, Taiwan

- Serve as Co-Principal Investigator to lead high-impact robotics projects and manage university-industry collaborations.
- Develop advanced digital twin and scene reconstruction pipelines for industrial automation.

Research Intern, Sony *Oct. 2023 – Mar. 2024*

Tokyo, Japan

- Conduct the research projects on Reinforcement Learning integrated with Large Language Models (LLMs).
- Collaborate with global teams to enhance agent guidance in complex environments.

Education

National Taiwan University (NTU) *Taipei, Taiwan*

Ph.D. in Computer Science and Information Engineering *Sep. 2019 – Jul. 2025*

- **GPA:** 4.22/4.3
- **Advisor:** Prof. Winston H. Hsu

National Taiwan Normal University (NTNU) *Taipei, Taiwan*

B.S. & M.S. in Computer Science and Information Engineering *Sep. 2013 – Jul. 2019*

- **GPA:** 3.74/4.3 (B.S.) & 4.3/4.3 (M.S.)

Research Experience

Research Focus: Developing reliable and safe autonomous systems to ensure AI efficiency and safety in complex real-world environments.

- Robotic Manipulation (AAAI 2022): Developed an attention-based policy for robots to learn multi-stage tasks from limited human demonstrations.
- System Safety (NeurIPS 2024): Proposed a monitoring framework to detect and prevent erroneous AI actions, ensuring high reliability in unpredictable environments.
- Complex Task Learning (ICLR 2025): Integrated LLM/VLM agents to improve AI's ability to follow long-horizon human instructions, achieving a 43% efficiency boost.

Skills

- **Programming:** C/C++, Python, C#, SQL
- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Weights&Biases
- **Cloud & Infrastructure:** AWS (EC2), Docker, Git, OpenMP
- **Robotics & Simulation:** ROS2, Isaac Sim, Pybullet, Coppeliasim

Academic Services

- **Area Chair:** ICLR 2026, NeurIPS 2026
- **Reviewer:** Regularly invited to review for top-tier AI conferences, including CVPR (Outstanding Reviewer, 2025), NeurIPS, ICML, ICLR, ICCV, ACL Rolling Review, and AAAI.

Selected Awards

- **Research & Dissertation Excellence:**
  - Hon Hai (Foxconn) Technology Award (2025) – *Top-tier industry research recognition*
  - IPPR Dissertation Awards (2025) – *National recognition for outstanding PhD research*
  - NeurIPS Scholar Award (2024) – *Prestigious award from a world-leading AI conference*
- **Scholarships & Memberships:**
  - NTU Presidential Award for Graduate Students (2025) – *Highest honor for graduate students at National Taiwan University*
  - Member of Phi Tau Phi (2025) – *Awarded for exceptional academic excellence*
  - NOVATEK PhD Scholarship (2022) – *Competitive industry-sponsored academic scholarship*
- **Technical Competitions:**
  - Computer Chess Competition (2016–2022): Awarded 16 medals across various national and international events

Selected Publications | citations: 520+ | h-index: 8 | Source: Google Scholar

- **VICtoR: Learning Hierarchical Vision-Instruction Correlation Rewards for Long-horizon Manipulation**
  ICLR 2025 (co-first author) | Top-tier Machine Learning conference
- **AED: Adaptable Error Detection for Few-shot Imitation Policy**
  NeurIPS 2024 | Premier venue for Machine Learning research.

- **Stage Conscious Attention Network (SCAN): A Demonstration-conditioned Policy for Few-shot Imitation**
  AAAI 2022 (co-first author) | Top-tier AI conference **(15% acceptance rate)**
- **Large Margin Mechanism and Pseudo Query Set on Cross-Domain Few-Shot Learning**
  CVPR Workshop Competition Report 2020 | citation: 20+
- **Other Contributions:**
  - Lead and co-authored works in IEEE TMM, IEEE TAI, ICCV, EMNLP, CoRL, and ICRA.